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NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

Supplementary Examinations - December 2025

21BT702 / 20BTC206 – CLINICAL RESEARCH AND DATA MANAGEMENT

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Give a detailed note on target identification and validation during drug discovery process.	10	L*2	1	2,11
b)	Describe the methods involved in pharmacovigilance.	10	L2	2	2,9
2.	Write an explanatory note on:				
i)	Personalized medicine	10	L1	1	2,11
ii)	Immunotoxicology	10	L1	2	2,9
3. a)	Explain in detail the IND registration of a new drug during drug discovery process.	10	L1	1	2,11
b)	Discuss about the different types of Adverse Drug Reactions with examples.	10	L2	2	2,9
Unit-II					
4. a)	Describe clinical trial study design with reference to cross-sectional studies.	10	L2	3	2,3,9
b)	What is clinical data management? Explain the stages, process and phases involved in clinical data management.	10	L2	4	2,8
5. a)	Briefly explain phase 3 and phase 4 clinical trial.	10	L2	3	2,3,9
b)	Illustrate the different phases of clinical trial studies involving cancer patients.	10	L3	4	2,8
6. a)	Explain clinical trial study design with reference to epidemiology studies.	10	L3	3	2,3,9
b)	Elaborate the role of contract research organization in clinical trials.	10	L1	4	2,8
Unit-III					
7. a)	Explain the role and responsibilities of an investigator in clinical trials.	10	L2	5	2,8
b)	Comment on the study data standards. Add a note on its significance.	10	L2	5	2,8
8. a)	Describe the roles and regulations of EMEA in clinical research.	10	L2	5	2,8
b)	Discuss the various principles involved in ICH-GCP guidelines.	10	L2	5	2,8

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
